

SAPTHAGIRI COLLEGE OF ENGINEERING
BANGALORE-560 057

(ACCREDITED BY NAAC WITH 'A' GRADE)



**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

(ACCREDITED BY NBA)

INTRODUCTION TO PYTHON PROGRAMMING
(BPLCK105B)

NAME OF THE STUDENT:

USN:

Sem.:

Section:



SAPTHAGIRI COLLEGE OF ENGINEERING

(Recognized by AICTE, New Delhi, Affiliated to VTU, Belagavi)

(An ISO 9001:2015 and 14001:2015 certified Institution)

#14/5, Chikkasandra, Hesaraghatta Main Road, Bengaluru - 560057

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Vision:

- To create globally competent Electrical and Electronics Engineers who can contribute to the growth of the nation and serve the society.

Mission:

M1: To impart student centric quality education

M2: To nurture the talents and impart moral values to the students.

M3: To keep abreast the technical knowledge among students and faculty with industry–academia interaction.

M4: To enrich research and innovation methods in students

PROGRAM OUTCOMES (POs):

Upon graduating from the program (B.E. – Electrical and Electronics Engineering), the students are able to,

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions for complex problems.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs):

The graduates of Electrical and Electronics Engineering Program should be able to attain the following at the time of graduation.

PSO 1: Apply fundamental knowledge to identify, formulate, design and investigate various problems of Electrical Machines, Power Electronics, Micro-Controller, Control System, Power System, High Voltage Engineering.

PSO 2: Apply modern software tools for design, simulation and analysis of electrical and electronics systems, engage in lifelong learning and to successfully adapt in multidisciplinary environments.

PSO 3: Solve ethically and professionally various Electrical and Electronics Engineering problems in societal and environmental context.

COURSE OUTCOMES:

At the end of this course, the students will be able to

CO1	Demonstrate proficiency in handling loops and creation of functions.
CO2	Identify the methods to create and manipulate lists, tuples and dictionaries.
CO3	Identify the methods to create and manipulate lists, tuples and dictionaries.
CO4	Interpret the concepts of Object-Oriented Programming as used in Python.

LIST OF PROGRAM:

- 1.a Develop a program to read the student details like Name, USN, and Marks in three subjects. Display the student details, total marks and percentage with suitable messages.
b. Develop a program to read the name and year of birth of a person. Display whether the person is a senior citizen or not.

- 2.a. Develop a program to generate Fibonacci sequence of length (N). Read N from the console.
b. Write a function to calculate factorial of a number. Develop a program to compute binomial coefficient (Given N and R).

- 3.Read N numbers from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable messages.

- 4.Read a multi-digit number (as chars) from the console. Develop a program to print the frequency of each digit with suitable message.

- 5.Develop a program to print 10 most frequently appearing words in a text file. [Hint: Use dictionary with distinct words and their frequency of occurrences. Sort the dictionary in the reverse order of frequency and display dictionary slice of first 10 items].

- 6.Develop a program to sort the contents of a text file and write the sorted contents into a separate text file. [Hint: Use string methods strip(), len(), list methods sort(), append(), and file methods open(), readlines(), and write()].

- 7.Develop a program to backing Up a given Folder (Folder in a current working directory) into a ZIP File by using relevant modules and suitable methods.

- 8.Write a function named DivExp which takes TWO parameters a, b and returns a value c ($c=a/b$). Write suitable assertion for $a>0$ in function DivExp and raise an exception for when $b=0$. Develop a suitable program which reads two values from the console and calls a function DivExp.

9. Define a function which takes TWO objects representing complex numbers and returns new complex number with a addition of two complex numbers. Define a suitable class 'Complex' to represent the complex number. Develop a program to read N ($N \geq 2$) complex numbers and to compute the addition of N complex numbers.

10. Develop a program that uses class Student which prompts the user to enter marks in three subjects and calculates total marks, percentage and displays the score card details. [Hint: Use list to store the marks in three subjects and total marks. Use `__init__()` method to initialize name, USN and the lists to store marks and total, Use `getMarks()` method to read marks into the list, and `display()` method to display the score card details.]

Expt. No.: 1**Date:**

AIM: - a) Develop a program to read the student details like Name, USN, and Marks in three subjects. Display the student details, total marks and percentage with suitable messages.

b) Develop a program to read the name and year of birth of a person. Display whether the person is a senior citizen or not.

PYTHON PROGRAM: -

- a. Develop a program to read the student details like Name, USN, and Marks in three subjects. Display the student details, total marks and percentage with suitable messages.

Program:

```
studentname = input('Enter the Name of the student: ')
usn = input('Enter the USN of the student: ')
m1 = int(input('Enter the mark in first subject: '))
m2 = int(input('Enter the mark in second subject: '))
m3 = int(input('Enter the mark in third subject: '))
total = m1+m2+m3
percentage = round((total/300)*100)

print('\nName of the Student: ', studentname)
print('USN: ', usn)
print('Mark in Subject 1: ', m1)
print('Mark in Subject 2: ', m2)
print('Mark in Subject 3: ', m3)
print('Total Score = ', total)
print('Percentage = ', percentage)
```



```
if percentage >=90:  
    print('First Class Exemplary.')
```

```
elif percentage >=75 and percentage <90:  
    print('First Class with Distinction.')
```

```
elif percentage >=60 and percentage <75:  
    print('First Class.')
```

```
elif percentage >=35 and percentage <60:  
    print('Second Class.')
```

```
else:  
    print('Fail.')
```

Output:

Enter the Name of the student: Vinay R
Enter the USN of the student: 1HK22EC0127
Enter the mark in first subject: 90
Enter the mark in second subject: 95
Enter the mark in third subject: 99

Name of the Student: Vinay R
USN: 1HK22EC0127
Mark in Subject 1: 90
Mark in Subject 2: 95
Mark in Subject 3: 99
Total Score = 284
Percentage = 95
First Class Exemplary.

b. Develop a program to read the name and year of birth of a person. Display whether the person is a senior citizen or not.

Program:

```
personname = input('Enter the Name of the person: ')
yearofbirth = int(input('Enter the Year of Birth: '))
currentyear = 2023
age = currentyear - yearofbirth

print("\nName of the person: ", personname)
print('Year of Birth of the person: ', yearofbirth)
print('Age of the person: ', age)

if age >= 60:
    print('The person is a Senior Citizen.')
else:
    print('The person is not a Senior Citizen.')
```

Output:

```
Enter the Name of the person: Saba
Enter the Year of Birth: 1986
```

```
Name of the person: Saba
Year of Birth of the person: 1986
Age of the person: 37
The person is not a Senior Citizen.
```

Expt. No.: 2**Date:****AIM:** - a) Develop a program to generate Fibonacci sequence of length (N).

Read N from the console.

b) Write a function to calculate factorial of a number. Develop a program to compute binomial coefficient (Given N and R).

PYTHON PROGRAM: -

a) Develop a program to generate Fibonacci sequence of length (N). Read N from the console.

Program:

```
f0 = 0
f1 = 1
N = int(input('Enter the Length of required Fibonacci Sequence: '))

if N<=0:
    print('Enter Positive Integer value: ')
else:
    i=0
    print('Fibonacci Sequence for N = '+ str(N) + ' is: ')
    while i<N:
        print(f0)
        fth = f0 + f1
        f0 = f1
        f1 = fth
        i += 1
```

Output:

Enter the Length of required Fibonacci Sequence: 10

Fibonacci Sequence for N = 10 is:

```
0
1
1
2
3
5
8
13
21
34
```

b) Write a function to calculate factorial of a number. Develop a program to compute binomial coefficient (Given N and R).

Program:

```
def factorial(N):
    if N<0:
        print('Enter the positive integer')
    elif N==0:
        print('The Factorial of 0 is 1')
    elif N==1:
        print('The Factorial of 1 is 1')
    else:
        i=1
        fl=1
        while i<=N:
            fth=fl*i
            fl=fth
            i+=1
        return fth

N=int(input('Enter the Number to find factorial: '))
fact = factorial(N)
print('The Factorial of ' +str(N)+ ' is ', fact)
```

Output:

Enter the Number to find factorial: 5
The Factorial of 5 is 120

Binomial coefficient:**Program:**

```
def factorial(N):
    if N<0:
        print('Enter the positive integer')
    elif N==0:
        print('The Factorial of 0 is 1')
    elif N==1:
        print('The Factorial of 1 is 1')
    else:
        i=1
        fl=1
        while i<=N:
            fth=fl*i
            fl=fth
            i+=1
        return fth
```

```
N=int(input('Enter the value of N: '))
R=int(input('Enter the value of R '))
factN = factorial(N)
factR = factorial(R)
binomial = factN / (factR*factorial(N-R))
print('The Binomial coefficient for N= ' +str(N)+ ' and R= '+str(R)+ ' is ', binomial)
```

Output:

Enter the value of N: 7

Enter the value of R 3

The Binomial coefficient for N= 7 and R= 3 is 35.0

Expt. No.: 3

Date:

AIM: - Read N numbers from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable messages.

PYTHON PROGRAM: -

<i>Categories</i>	<i>Score</i>	<i>Reliability level</i>
High dispersion	1.00 a 1.50	Low reliability
Low dispersion	0.50 a 0.99	Reliable
Very low dispersion	0.00 a 0.49	High reliability

```
N=int(input('Enter the number of elements: '))
```

```
lis=[]
for i in range(N):
    lis.append(int(input('Enter the element: ')))
```

```
mean = round(sum(lis)/N,2)
print("\nMean is ',mean)
```

```
lis_mean=[]
for i in range(N):
    lis_mean.append((lis[i] - mean)**2)
```

```
variance = round(sum(lis_mean)/N,2)
print('Variance is ', variance)
```

```
SD=round(variance**(1/2),2)
print('Standard deviation is ', SD)
if SD>=1.5:
    print('High Dispersion, Low Reliability.')
else:
    print('Low Dispersion, Reliable.')
```

Output:

Enter the number of elements: 8

Enter the element: 10

Enter the element: 12

Enter the element: 23

Enter the element: 23

Enter the element: 16

Enter the element: 23

Enter the element: 21

Enter the element: 16

Mean is 18.0

Variance is 24.0

Standard deviation is 4.9

High Dispersion, Low Reliability.

Expt. No.: 4**Date:**

AIM: - Read a multi-digit number (as chars) from the console. Develop a program to print the frequency of each digit with suitable message.

PYTHON PROGRAM: -

```
import pprint

message = input('Enter your input: ')

count = {}

for character in message:

    count.setdefault(character, 0)

    count[character] = count[character] + 1

print('Frequency of words appeared in the Sentence or paragraph\n')

pprint.pprint(count)

print()

for k,v in count.items():

    print('Character \''+str(k)+'\' appeared ' + str(v)+ ' times')
```


Output:

Enter your input:

292749028574908279082574258979529847209846987620234456025657260974502874597264652

Frequency of words appeared in the Sentence or paragraph

```
{'0': 8,  
'2': 15,  
'3': 1,  
'4': 10,  
'5': 10,  
'6': 7,  
'7': 11,  
'8': 8,  
'9': 11}
```

Character '2' appeared 15 times

Character '9' appeared 11 times

Character '7' appeared 11 times

Character '4' appeared 10 times

Character '0' appeared 8 times

Character '8' appeared 8 times

Character '5' appeared 10 times

Character '6' appeared 7 times

Character '3' appeared 1 times

Expt. No.: 5**Date:**

AIM: - Develop a program to print 10 most frequently appearing words in a text file. [Hint: Use dictionary with distinct words and their frequency of occurrences. Sort the dictionary in the reverse order of frequency and display dictionary slice of first 10 items].

Before starting the program save some content in .txt file, and use the path for the same file in
message1=open(_____)

PYTHON PROGRAM: -

```
import pprint
message1 =open(r"C:\Users\Sathish\Desktop\bluetooth.txt")
message = message1.read()
message1.close()
print('The content in the file is: \n\n', message)
count = {}
for character in message.split():
    count.setdefault(character, 0)
    count[character] = count[character] + 1
print("\nFrequency of words appeared in the Sentence or paragraph\n")
pprint.pprint(count)
print()
sortedcount = sorted(count.items(), key=lambda x:x[1], reverse=True)
converttodict = dict(sortedcount)
print("\nSorted in descending order:\n\n',converttodict)
import itertools
sliceddict = dict(itertools.islice(converttodict.items(), 10))
print("\nTen Most frequently appeared word in a paragraph :\n\n',sliceddict)
```

Output:

The content in the file is:

Bluetooth technology is a high speed, low powered wireless technology that is used to transmit and receive data serially. The Bluetooth transceivers consist of many devices such as mobile phones, computers, and other electronic devices. Bluetooth technology is one of the best seminar topics for electronics and communication students. In the embedded system many of electronic project applications, controlling by Bluetooth technology. The Bluetooth technology gets second place in Seminar topics for electronics and communication.

Frequency of words appeared in the Sentence or paragraph

```
{'Bluetooth': 5,  
 'In': 1,  
 'Seminar': 1,  
 'The': 2,  
 'a': 1,  
 'and': 4,  
 'applications.': 1,  
 'as': 1,  
 'best': 1,  
 'by': 1,  
 'communication': 1,  
 'communication.': 1,  
 'computers.': 1,  
 'consist': 1,  
 'controlling': 1,  
 'data': 1,  
 'devices': 1,  
 'devices.': 1,  
 'electronic': 2,  
 'electronics': 2,  
 'embedded': 1,  
 'for': 2,  
 'gets': 1,  
 'high': 1,  
 'in': 1,  
 'is': 3,  
 'low': 1,  
 'many': 2,  
 'mobile': 1,
```

'of': 3,
'one': 1,
'other': 1,
'phones.': 1,
'place': 1,
'powered': 1,
'project': 1,
'receive': 1,
'second': 1,
'seminar': 1,
'serially.': 1,
'speed.': 1,
'students.': 1,
'such': 1,
'system': 1,
'technology': 4,
'technology.': 1,
'that': 1,
'the': 2,
'to': 1,
'topics': 2,
'transceivers': 1,
'transmit': 1,
'used': 1,
'wireless': 1}

Sorted in descending order:

{'Bluetooth': 5, 'technology': 4, 'and': 4, 'is': 3, 'of': 3, 'The': 2, 'many': 2, 'electronic': 2, 'the': 2, 'topics': 2, 'for': 2, 'electronics': 2, 'a': 1, 'high': 1, 'speed.': 1, 'low': 1, 'powered': 1, 'wireless': 1, 'that': 1, 'used': 1, 'to': 1, 'transmit': 1, 'receive': 1, 'data': 1, 'serially.': 1, 'transceivers': 1, 'consist': 1, 'devices': 1, 'such': 1, 'as': 1, 'mobile': 1, 'phones.': 1, 'computers.': 1, 'other': 1, 'devices.': 1, 'one': 1, 'best': 1, 'seminar': 1, 'communication': 1, 'students.': 1, 'In': 1, 'embedded': 1, 'system': 1, 'project': 1, 'applications.': 1,

```
'controlling': 1, 'by': 1, 'technology.': 1, 'gets': 1, 'second': 1, 'place': 1, 'in': 1, 'Seminar': 1, 'communication.': 1}
```

Ten Most frequently appeared word in a paragraph :

```
{'Bluetooth': 5, 'technology': 4, 'and': 4, 'is': 3, 'of': 3, 'The': 2, 'many': 2, 'electronic': 2, 'the': 2, 'topics': 2}
```

Expt. No.: 6**Date:**

AIM: - Develop a program to sort the contents of a text file and write the sorted contents into a separate text file. [Hint: Use string methods strip(), len(), list methods sort(), append(), and file methods open(), readlines(), and write()].

PYTHON PROGRAM: -

```
namelist=open(r'C:\Users\Sathish\Desktop\namelist.txt')
namelist1=namelist.readlines()
namelist.close()
print("\n\n content available of the file is:\n\n",namelist1)
namelist1.sort()
sortedlist2=open(r'C:\Users\Sathish\Desktop\sortedlist.txt','w')
sortedlist2.writelines(namelist1)
sortedlist2.close()
print("\n\n Sorted content is :\n\n",namelist1)
```

Output:

content available of the file is:

```
['PRUTVIRAJ SANTOSHAKUMAR TUPPAD\n', 'PRUTHVI BT\n', 'PRIYANSHU KUMAR\n',
'PRIYA GT\n', 'PRAMOD SIDDAPPA PUJAR\n', 'PRADNESH R\n', 'PRADEEP.SS\n', 'POOJITHA
S\n', 'PAVANI M K\n', 'PARVEEN\n', 'PALLAVI N K\n', 'NUFAIZ ZULFIKAR\n', 'NITHIN
SADASHIV SHIPURE\n', 'NITHIN S\n', 'NISHANTH GOWDA VN\n', 'NISARGA K P\n',
'NISARGA CHIVATE\n', 'NISARGA C\n', 'NIRIKSHA C N\n', 'NIHAR FATHIMA\n', 'NAYAN
GOWDA R\n', 'NAWAZ AHMED\n', 'NAVYA SHREE R\n', 'NASIR KHAN\n', 'NARLA
VARSHITHA\n', 'NAGIREDDY NANDINI\n', 'MUTTURAJ\n', 'MUSARATH MUNTAHA\n',
'MUHAMMAD FARHAN\n', 'MOULYASHREE K M\n', 'MONISHA C\n', 'MONIKA B R\n',
'MOHAMMED UMAR ALI\n', 'MOHAMMED SINAN\n', 'MOHAMMED MAAZ H\n',
'MOHAMMED KHIZER PASHA\n', 'MOHAMMED KHIZER M D\n', 'MOHAMMED FARHAN\n',
'MOHAMMED BILAL PASHA\n', 'MOHAMMED ARSHAD\n', 'MOHAMMAD YAHYA\n',
'MELWIN JAIN\n', 'MEGHANA T V\n', 'MD AASIM AHMED\n', 'MARIA SWETHA A\n',
'MANDAPATI ANANYA\n', 'MAHASHWETHA G\n', 'M SATHYA SAI VENKATA
RANGAIAH\n', 'M JAGAN MOHAN REDDY\n', 'LIKITHA R\n', 'LIKHITH GOWDA S C\n',
'KUMAR K N\n', 'KULSUM HUZAIFA\n', 'KULADEEP GC\n', 'KOMAL\n', 'KOLLURI
AKANKSHA\n', 'KIRAN M N\n', 'KIRAN KUMAR N\n', 'KIRAN KUMAR GOUDAR \n', 'KIRAN
K S\n', 'KEERTHANA J\n', 'KEDARALINGA K S\n', 'KANCHANA H GOWDA']
```

Sorted content is :

```
['KANCHANA H GOWDA', 'KEDARALINGA K S\n', 'KEERTHANA J\n', 'KIRAN K S\n',
'KIRAN KUMAR GOUDAR \n', 'KIRAN KUMAR N\n', 'KIRAN M N\n', 'KOLLURI
AKANKSHA\n', 'KOMAL\n', 'KULADEEP GC\n', 'KULSUM HUZAIFA\n', 'KUMAR K N\n',
'LIKHITH GOWDA S C\n', 'LIKITHA R\n', 'M JAGAN MOHAN REDDY\n', 'M SATHYA SAI
VENKATA RANGAIAH\n', 'MAHASHWETHA G\n', 'MANDAPATI ANANYA\n', 'MARIA
```

SWETHA A\n', 'MD AASIM AHMED\n', 'MEGHANA T V\n', 'MELWIN JAIN\n', 'MOHAMMAD YAHYA\n', 'MOHAMMED ARSHAD\n', 'MOHAMMED BILAL PASHA\n', 'MOHAMMED FARHAN\n', 'MOHAMMED KHIZER M D\n', 'MOHAMMED KHIZER PASHA\n', 'MOHAMMED MAAZ H\n', 'MOHAMMED SINAN\n', 'MOHAMMED UMAR ALI\n', 'MONIKA B R\n', 'MONISHA C\n', 'MOULYASHREE K M\n', 'MUHAMMAD FARHAN\n', 'MUSARATH MUNTAHA\n', 'MUTTURAJ\n', 'NAGIREDDY NANDINI\n', 'NARLA VARSHITHA\n', 'NASIR KHAN\n', 'NAVYA SHREE R\n', 'NAWAZ AHMED\n', 'NAYAN GOWDA R\n', 'NIHAR FATHIMA\n', 'NIRIKSHA C N\n', 'NISARGA C\n', 'NISARGA CHIVATE\n', 'NISARGA K P\n', 'NISHANTH GOWDA VN\n', 'NITHIN S\n', 'NITHIN SADASHIV SHIPURE\n', 'NUFAIZ ZULFIKAR\n', 'PALLAVI N K\n', 'PARVEEN\n', 'PAVANI M K\n', 'POOJITHA S\n', 'PRADEEP.SS\n', 'PRADNESH R\n', 'PRAMOD SIDDAPPA PUJAR\n', 'PRIYA GT\n', 'PRIYANSHU KUMAR\n', 'PRUTHVI BT\n', 'PRUTVIRAJ SANTOSHAKUMAR TUPPAD\n']

Expt. No.: 7**Date:**

AIM: - Develop a program to backing Up a given Folder (Folder in a current working directory) into a ZIP File by using relevant modules and suitable methods.

PYTHON PROGRAM: -

```
import zipfile, os
def backupToZip(folder):
    folder = os.path.abspath(folder)
    number = 1
    while True:
        zipFilename = os.path.basename(folder) + '_' + str(number) + '.zip'
        if not os.path.exists(zipFilename):
            break
        number = number + 1
    print('Creating %s...' % (zipFilename))
    backupZip = zipfile.ZipFile(zipFilename, 'w')
    for foldername, subfolders, filenames in os.walk(folder):
        print('Adding files in %s...' % (foldername))
        backupZip.write(foldername)
        for filename in filenames:
            newBase = os.path.basename(folder)
            if filename.startswith(newBase) and filename.endswith('.zip'):
                continue
            backupZip.write(os.path.join(foldername, filename))
    backupZip.close()
    print('Done.')

# Main Program
backupToZip(r'C:\Users\Sathish\Desktop\bb')
```

Output:

Done.

Expt. No.: 8**Date:**

AIM: - Write a function named DivExp which takes TWO parameters a, b and returns a value c ($c=a/b$). Write suitable assertion for $a>0$ in function DivExp and raise an exception for when $b=0$. Develop a suitable program which reads two values from the console and calls a function DivExp.

PYTHON PROGRAM: -

```
def DivExp(a,b):
    try:
        c=a/b
    except ZeroDivisionError:
        print('Error: Division by zero')
    else:
        return c

a = int(input('Enter the Divident value of \'a\': '))
b = int(input('Enter the Divisor value of \'b\': '))
d=DivExp(a,b)

if d is not None:
    print('The result of division is: ', d)
else:
    print('The result of division is: infinity')
```

Output:

```
Enter the Divident value of 'a': 10
Enter the Divisor value of 'b': 0
Error: Division by zero
The result of division is: infinity
```

Expt. No.: 9**Date:**

AIM: - Define a function which takes TWO objects representing complex numbers and returns new complex number with a addition of two complex numbers. Define a suitable class 'Complex' to represent the complex number. Develop a program to read N ($N \geq 2$) complex numbers and to compute the addition of N complex numbers.

PYTHON PROGRAM: -

```
class Complex:
    def __init__(self, real, imag):
        self.real = real
        self.imag = imag

    def __add__(self, other):
        return Complex(self.real + other.real, self.imag + other.imag)

    def __str__(self):
        if self.imag >= 0:
            return f'{self.real}+{self.imag}i'
        else:
            return f'{self.real}{self.imag}i'

def add_complex_numbers(num1, num2):
    return num1 + num2

n = int(input("Enter the number of complex numbers: "))

complex_numbers = []
for i in range(n):
    real, imag = input(f'Enter complex number {i+1}: ').split()
    complex_numbers.append(Complex(float(real), float(imag)))

result = complex_numbers[0]
for i in range(1, n):
    result = add_complex_numbers(result, complex_numbers[i])

print("Sum of the complex numbers is:", result)
```

Output:

```
Enter the number of complex numbers: 2
Enter complex number 1: 1 2
Enter complex number 2: 2 3
Sum of the complex numbers is: 3.0+5.0i
```

Expt. No.: 10**Date:**

AIM: - Develop a program that uses class Student which prompts the user to enter marks in three subjects and calculates total marks, percentage and displays the score card details. [Hint: Use list to store the marks in three subjects and total marks. Use `__init__()` method to initialize name, USN and the lists to store marks and total, Use `getMarks()` method to read marks into the list, and `display()` method to display the score card details.]

PYTHON PROGRAM: -

```
class Student:
    def __init__(self, name, usn):
        self.name = name
        self.usn = usn
        self.marks = []
        self.total = 0

    def getMarks(self):
        for i in range(3):
            marks = float(input(f'Enter marks for subject {i+1}: '))
            self.marks.append(marks)
            self.total += marks

    def display(self):
        print("Name:", self.name)
        print("USN:", self.usn)
        print("Marks:")
        for i in range(3):
            print(f'\tSubject {i+1}: {self.marks[i]}')
        print("Total:", self.total)
        print("Percentage:", round((self.total / 3), 2))

# Main program
name = input("Enter name: ")
usn = input("Enter USN: ")
```

```
student = Student(name, usn)
student.getMarks()
student.display()
```

Output:

```
Enter name: Kanchana
Enter USN: 1HK22EC065
Enter marks for subject 1: 98
Enter marks for subject 2: 76
Enter marks for subject 3: 56
Name: Kanchana
USN: 1HK22EC065
Marks:
    Subject 1: 98.0
    Subject 2: 76.0
    Subject 3: 56.0
Total: 230.0
Percentage: 76.67
```